

Ji Woong John Park

Boston, MA • US Citizen • johnpark0528@gmail.com • Phone: (484)716-6292 • [GitHub](#)

Education

Pennsylvania State University

Bachelor of Science 2024

Computer Science

Mathematics Minor

GPA: 3.69/4.0

Relevant Coursework:

- └ Systems Programming in C
- └ Theory of Computation
- └ Computer Networking
- └ Intro to Neural Networks
- └ Java Programming
- └ Big Data Computation
- └ Database Management
- └ Operating Systems Design
- └ Math Theory of Games
- └ Data Struct. & Algorithms

Skills

Python • TypeScript • C • SQL •
React Native • JavaScript • Git
• Pandas • C# • Docker •
Vb.Net • AWS • Power BI •
HTML/CSS • Angular

Languages

English (*Native*)
Korean (*Proficient*)

Activities & Interests

- └ Website Developer for Nittany Data Labs
- └ Korean Language Tutor
- └ Guiding Tours
- └ Chess

Awards and Honors

- └ 1st Place Optum UHC B&E Mini Hackathon
- └ Dean's list: SP21, FA21, SP22, FA22, FA23, SP24

Work Experience

Optum/UnitedHealth Group

Full Stack Engineer

Feb 2025-Current

- Designed event-driven processors to consume events and developed a RESTful API for cross-platform core communication through Fusion Events Gateway.
- Fixed a SQL parameter overflow bug (2,100+ limit) causing encounter failures and delivered client portal UI enhancements.
- Introduced new table pattern for transactional consistency across multiple entity types, refactoring legacy inline publishing into a scalable, dataset-persisted model with event tree traversal support.
- Built Angular client portal UI updates resolved a production WebApp bugs.

Full Stack App

Jun 2023-Aug 2023

- Worked through full stack development of end-to-end app for seamless business interactions between client and company in CI/CD and Agile workflow.
- Tech Stack: TypeScript, React Native, and Figma to build dashboard view of patient data within a modern UI/UX enhancing user experience. Then connected back-end data with internal APIs for data retrieval.

Back End Data Sourcing

Jun 2022-Aug 2022

- Built prediction algorithm that retrieves missing Covid-19 Vaccination information from U.S. state registries using patient location.
- Data analysis with Pandas and Python. Data parsing using SQL in DBeaver for big data management. Worked in Agile Scrum Teams.

Selected Projects

Handling Chaos: Multi-threading handler in C • CMPSC 473 OS Design

- Programmed in C to organize multi-threading in a CPU processor to avoid race conditions, deadlocks, and starvation.

FPGA Magic: Mini-CPU Architecture FPGA • CMPEN 331 CPU Design

- Created a mini-CPU programmed in MIPS on Vivado that executes machine code.
- Tested with Verilog to fetch, execute, and write data from registers and memory.

App Development: Android App • CodePath Android Development

- Constructed an Android app "Recishop" that transforms recipes into curated shopping lists, simplifying grocery planning and enhancing user experience.
- Other projects include Instagram clone, Twitter clone, Movie Rating app.

Leadership

Association for Computing Machinery President

Nov 2022-Nov 2023

- Direct and facilitate activities that provide technical opportunities to peer students. Manage 14 club executives on a weekly basis.

Research

- Research the implementation of Natural Language Processing neural network to detect word pattern and sentence structure to detect Twitter users who show depressive behavior on social media posts.
- Extensive research in publications regarding Long Short-Term Memory, Graphical Neural Networks, and Lexical Analysis.